

STAT 390 Capstone

Week 7: Final Artifact Assembly and Presentation

Goal: Present a complete, honest, evidence-backed AutoResearch project.



Week 7 Mindset

Stop searching. Start
confirming.

This week is not about trying one more idea. It is about showing that your final result is **credible**, your story is **clear**, and your research loop **taught us something**.



What This Week Must Achieve

By the end of this week, each student should have all five of the following in place:



Final Candidate

A final candidate method or workflow selected and locked



Locked Results

A locked final results table — no more edits after this point



Stability Evidence

Evidence that the result is stable and not a one-off run



Clear Story

A clear, coherent final project story ready to present



Agent Reflection

A reflection on what the agent did well and where it fell short

Final Presentation Rules

Hard Limits

- Maximum 6 minutes
- Maximum 10 slides
- No live coding

What to Avoid

- No long background section — get to the point fast
- No new experiments after final test evaluation

✓ **Recommended target:** 5 minutes, 7–8 slides.



Required Final Submission

Your final submission should make the project **reproducible**, **understandable**, and **honest**. Please submit all of the following:

1

Final Slides

Your polished presentation deck, ready to deliver

2

Final Report

Written documentation of your full research process (4 pages using the NeurIPS template)

3

Cleaned Repository

Organized, commented code that others can navigate

4

Experiment Archive

Full record of all experiments run during the project

5

Final Results Table

Locked table of all key metrics and outcomes

6

Reflection Memo

Honest assessment of agent performance and your learning

What the Presentation Must Answer

Your final presentation must address each of these five core questions clearly and in order:

01

From Topic to Contract

How did your original topic become an AutoResearch contract?

02

Directions Explored

What directions did the agent explore throughout the loop?

03

Stable Value

Which directions produced stable, reproducible value?

04

Noise and Failure

Which directions produced noise, dead ends, or outright failure?

05

Limits of the Loop

What are the fundamental limits of your AutoResearch loop?

Suggested 6-Minute Talk Structure

Use this pacing guide to structure your delivery. Each segment has a clear purpose — don't let any one section run over.



Recommended Slide Deck Structure

A strong final deck should follow this structure. Optional additions like a demo screenshot or appendix may be included after slide 8.

1

Title + Claim

One-sentence claim up front

2

Topic → AutoResearch Formulation

How the research question was formalized

3

Data & Baseline

Metric definition and starting point

4

Loop Design

Agent architecture and iteration strategy

5

Experiment Trace

Key experiments and what they revealed

6

Final Result

Result and stability check evidence

7

Worked vs. Failed

Honest breakdown of outcomes

8

Reflection

Limits and lessons learned

Credibility Check

Your result should not be a one-off lucky run.

Be ready to answer each of these questions about your methodology and evaluation process:

Test Set Integrity

Was the final test set opened **only once**? Any peeking invalidates your result.

No Leakage

Did the test result influence further tuning? If yes, your evaluation is compromised.

Fixed Evaluator

Did the evaluator remain fixed throughout? Changing it mid-project undermines comparisons.

Stability Verified

Was the result repeated or checked for stability across multiple runs?

Evidence for Claim

What concrete evidence supports your final claim? Be specific and direct.

What Counts as a Strong Project

- ✓ A strong project does **not** need to be a breakthrough. Rigor and honesty matter more than novelty.

Any of the following outcomes, executed with discipline and transparency, constitutes a strong capstone project:

Credible Improvement

A measurable, reproducible improvement over a well-defined baseline

Rigorous Benchmark

A careful, well-documented benchmarking study with clear methodology

Reproducible Pipeline

A clean, reusable workflow that others can run and build upon

Useful Workflow

A practical AutoResearch loop that solves a real problem reliably

Negative Result

A convincing demonstration that a promising approach does not work

Failure Analysis

A clear, insightful analysis of why a method failed and what that reveals



Reflection Questions

Your reflection memo should go beyond reporting a score. The goal is to genuinely understand what agent-assisted research looks like in practice. Answer each of the following:



What did the agent do well?

Identify specific tasks or phases where the agent added clear, measurable value to your research process.



What did the agent do poorly?

Be honest about where the agent struggled, made errors, or required significant correction from you.



What required human judgment?

Which decisions could not be delegated to the agent? Where was your own expertise irreplaceable?



How would you redesign the loop?

If you restarted the project from scratch, what would you change about the AutoResearch loop design?

Final Takeaway

A successful final presentation demonstrates all five of the following qualities together:

Clear Research Question

A well-defined, answerable question that drove the entire project

Disciplined Loop

A structured AutoResearch loop with principled iteration and evaluation

Credible Evidence

Results that are stable, reproducible, and honestly reported

Honest Failure Analysis

A transparent account of what did not work and why

Thoughtful Reflection

Genuine insight into human-agent collaboration and its limits

📌 **Final question:** What did you learn about doing research with AI agents?